MATH 101

Chapter 2 Homework

Equations and inequalities

Sections 2.1-2.2 and 2.4-2.7 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Give exact answers. Use a basic four-function calculator only if needed; show algebraic reasoning.

Equations and their graphs

1. Find the x- and y-intercepts, then sketch the line.

$2x + 3y = 12$

2. Solve. Check your solution.

$3(2x - 1) - 4 = 2(x + 5)$

3. Solve each equation; describe the full solution set.

a) $2(x - 3) + 5 = 2x - 1$

b) $3(x + 1) = 3x - 2$

4. State the excluded input, solve, and check in the original equation.

$\frac{x+1}{x-2} = 2$

Quadratic and other equations

5. Solve by factoring.

$x^2 - 5x - 14 = 0$

6. Solve by taking square roots.

$3(x - 2)^2 = 27$

7. Solve with the quadratic formula. Give exact answers.

$2x^2 - 3x - 1 = 0$

8. Solve over the complex numbers. State why there are no real solutions.

$x^2 + 6x + 13 = 0$

9. Solve. Check every candidate in the original equation.

$\sqrt{x + 6} = x$

10. Solve by treating the equation as a quadratic in x squared.

$x^4 - 5x^2 + 4 = 0$

11. Solve.

$$|2x - 1| = 7$$

Inequalities and reasoning

12. Solve and write the answer in interval notation.

$$5 - 2x \geq 11$$

13. Solve and sketch the solution on a number line.

$$-3 < 2x + 1 \leq 7$$

14. Solve and write the answer in interval notation.

$$|x - 2| < 5$$

15. Solve and write the answer in interval notation.

$$|x + 1| \geq 3$$

16. A student solves x(x - 4) = 0 by dividing by x and reports only x = 4. Identify the lost solution and explain which division was invalid.